

TASK 1:

The screenshot shows the Azure Portal interface for a Web App. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), Log stream, Resource visualizer, Deployment, Settings, Environment variables, and Configuration. The main content area displays the 'Overview' page for the Web App 'pa4-27100003'. It includes a search bar, action buttons (Browse, Stop, Swap, Restart, Delete, Refresh, Download publish profile, Reset publish profile, Share to mobile), and a 'View Cost' link. The 'Essentials' section shows key information: Resource group (rg-sp26-27100003), Status (Running), Location (UK West), Subscription (Microsoft Azure - Ali Khawaja), Subscription ID (67e93b84-fe08-452c-80ea-175d0a3eca56), Default domain (pa4-27100003.azurewebsites.net), App Service Plan (pa4-27100003 (B1: 1)), Operating System (Linux), and Health Check (Cannot fetch health check data. Please try again later.). Below this, the 'Properties' tab is active, showing details for the 'Web app' resource, including Name (pa4-27100003), Publishing model (Code), and Runtime Stack (Node - 22-lts).

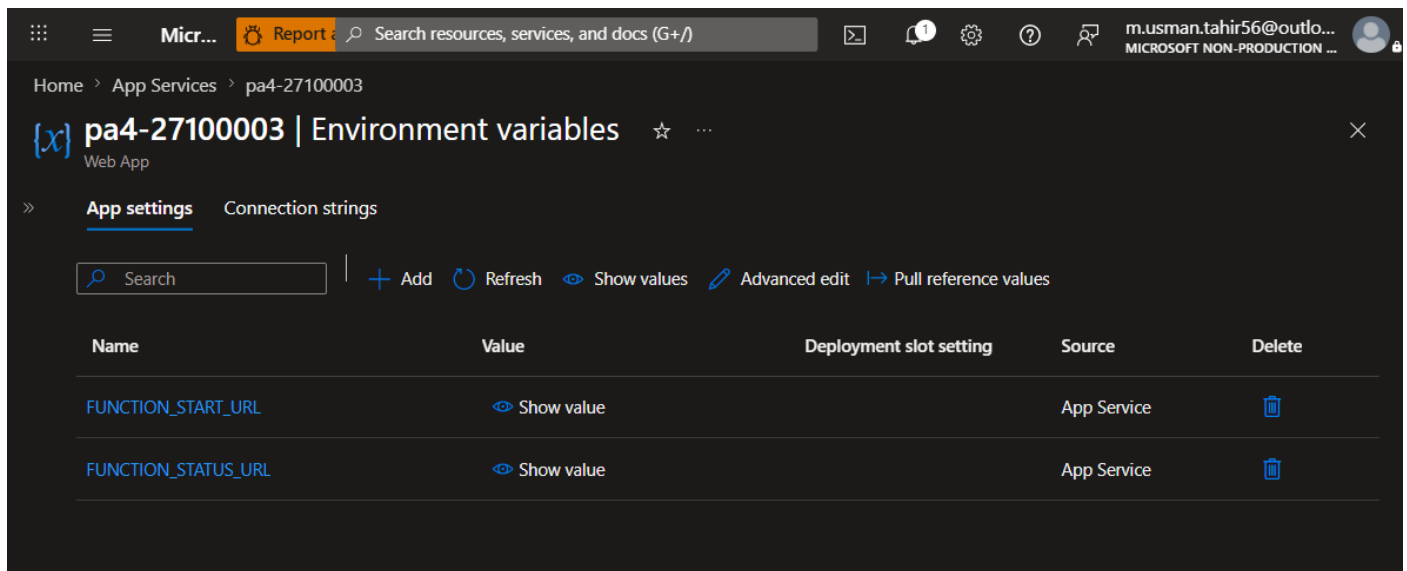
app service running

The screenshot shows the Azure Portal interface for the 'Deployment Center' of a Web App named 'pa4-27100003'. The left sidebar shows the navigation menu. The main content area displays the 'Deployment Center' page, which includes tabs for Settings, Containers (new), Logs, and FTPS Credentials. The 'Logs' tab is active, showing a table of deployment logs. The table has columns for Time, Deployment ID, Log source, Author, Status, and Message. A log entry is visible for 'Wednesday, May 6, 2026 (1)' at '5/6/2026, 07:51:34 PM' with Deployment ID 'ddce766', Log source 'Build/Deploy logs', Author 'usmantahirr', Status 'Failed', and Message 'Add or update the Azure App Service build and deployment workflow config'.

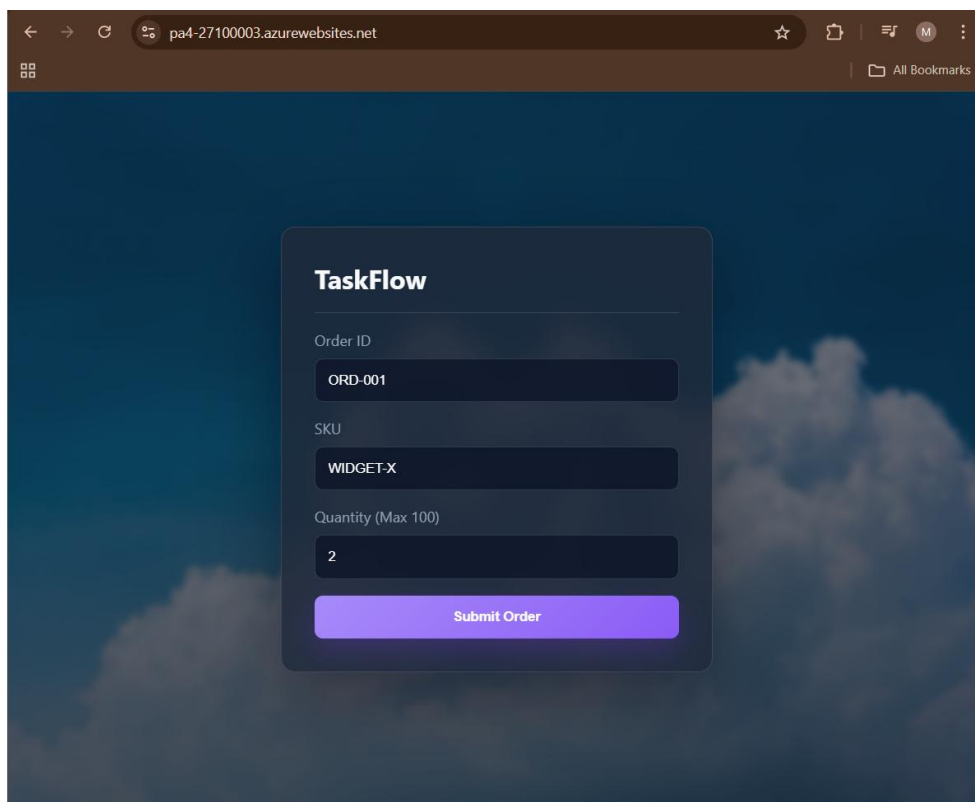
deployment center connected with my github

The screenshot shows the GitHub Actions workflow page for a repository named 'CS487-PA4'. The top navigation bar includes links for Code, Pull requests, Agents, Actions, Projects, Wiki, Security and quality, Insights, and Settings. The 'Actions' tab is active, showing a workflow named 'fix: deploy only webapp subfolder #2'. The workflow status is 'Success', triggered by a push event. The workflow details show a summary of the jobs: 'build' and 'deploy'. The 'build' job took 9s and the 'deploy' job took 36s. The workflow is currently running, and the status is green.

fixed the error of build: now its deployed, workflow is green

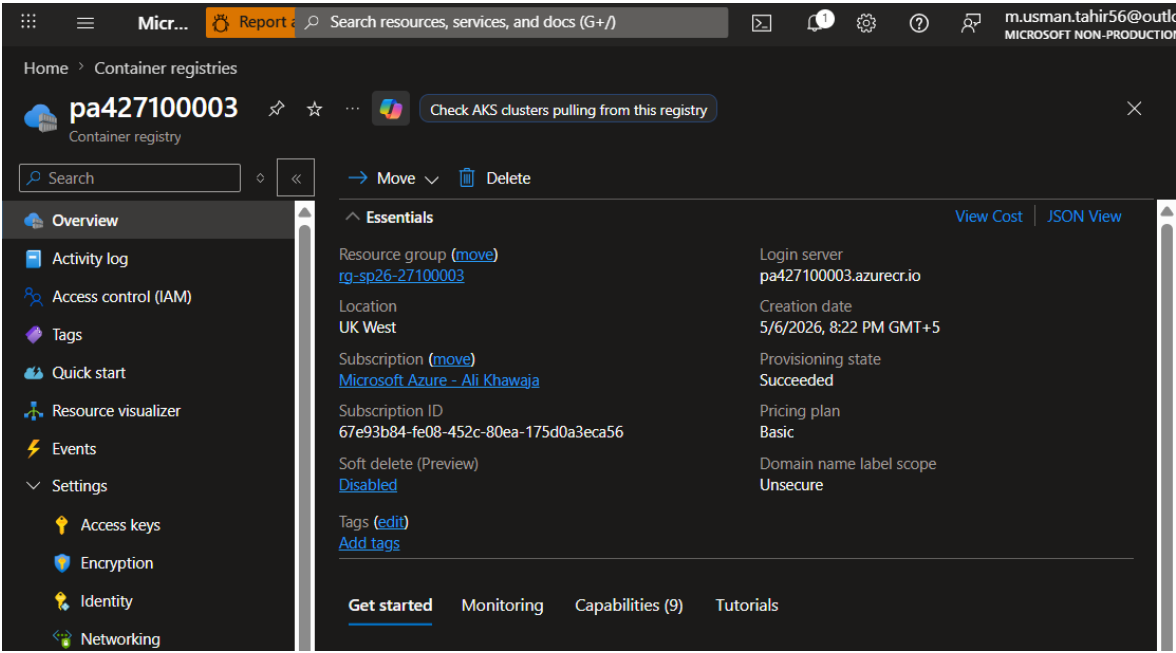


both FUNCTION_START_URL and FUNCTION_STATUS_URL listed



successful load of the dashboard

TASK 2:



overview of the pa427100003 in container registries

```
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> docker images
REPOSITORY                                TAG      IMAGE ID      CREATED        SIZE
func-app                                  latest   0393609d2294  32 seconds ago 1.95GB
report-job                                latest   5cef90212c9b  26 minutes ago 194MB
validate-api                              latest   b23343d585d6  28 minutes ago 149MB
crpa327100003.azurecr.io/pa3-27100003-app latest   2f8d3b996a32  3 weeks ago   152MB
pa3-app                                   latest   2f8d3b996a32  3 weeks ago   152MB
netcen-pa4-image                          latest   9728a47376fe  12 months ago 598MB
netcen-pa3-image                          latest   412322890572  12 months ago 615MB
<none>                                     <none>    ffd8b2b2efdb7 12 months ago 605MB
netcen-pa2-image                          latest   a0dc33bcd88a  14 months ago 626MB
netcen-pa1-image                          latest   b4af2afc554d  14 months ago 626MB
```

all 3 images created (top 3)

```
Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): y

StatusCode      : 200
StatusDescription : OK
Content          : {"valid":true,"reason":"ok","order_id":"LOCAL-1"}
RawContent       : HTTP/1.1 200 OK
                  Content-Length: 49
                  Content-Type: application/json
                  Date: Wed, 06 May 2026 15:56:44 GMT
                  Server: uvicorn

                  {"valid":true,"reason":"ok","order_id":"LOCAL-1"}
Forms           : {}
Headers         : [[Content-Length, 49], [Content-Type, application/json], [Date, Wed, 06 May 2026
                  15:56:44 GMT], [Server, uvicorn]]
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 49
```

terminal output showing StatusCode: 200 and the JSON response.

```
>> docker tag report-job:latest pa427100003.azurecr.io/report-job:v1
>> docker tag func-app:latest pa427100003.azurecr.io/func-app:v1
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> docker push pa427100003.azurecr.io/validate-api:v1
>> docker push pa427100003.azurecr.io/report-job:v1
>> docker push pa427100003.azurecr.io/func-app:v1
The push refers to repository [pa427100003.azurecr.io/validate-api]
e812d90f81e0: Pushed
558b83f0e2ff: Pushed
172e81dceb15: Pushed
e4b4e2c422df: Pushed
2360b28b4660: Pushed
248c30140986: Pushed
fa1aec823035: Pushed
6d7c150df58d: Pushed
v1: digest: sha256:a72d84caf63943641f07cb8397997198d4bb479761daa1ad08da936f839ed304 size: 1990
The push refers to repository [pa427100003.azurecr.io/report-job]
3a820128b360: Pushed
df7a2e54d6f7: Pushed
661dc515c9af: Pushed
e4b4e2c422df: Mounted from validate-api
2360b28b4660: Mounted from validate-api
248c30140986: Mounted from validate-api
fa1aec823035: Mounted from validate-api
6d7c150df58d: Mounted from validate-api
v1: digest: sha256:0c3db82c41b21cad37fd8ee4761468bc8325345b5434caa2372c1c1b54fd623b size: 1991
The push refers to repository [pa427100003.azurecr.io/func-app]
54bea61b5022: Pushed
faddf566e6b6: Pushed
ed8764339d1a: Pushed
6521412ea0a2: Pushed
bf9bcadb693e: Pushed
5f70bf18a086: Pushed
8ee04fca66f6: Pushed
56012687890a: Pushed
0f9961f81a08: Pushed
56fd19c1de72: Pushed
c62009188b5c: Pushed
8d4bbc4160d0: Pushed
07a72ddd5ffb: Pushed
5f0f98883fab: Pushed
8a3e3a5f0fef: Pushed
8a65bc77d8ca: Pushed
6d767d7d4483: Pushed
2694b17e7cb0: Pushed
2d5fd2ed1d7b: Pushed
14c1dc5943c0: Pushed
9b7261696dcc: Pushed
v1: digest: sha256:dc9ce66cd19b36f30f894ff8aac0aa1f1ebcb328b2c6c6ab9930a9f6b76a9d21 size: 4733
```

all 3 images pushed successfully

```
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> az acr repository list --name pa427100003 --output table
Result
-----
func-app
report-job
validate-api
```

output of az acr repo list

The screenshot shows the Azure portal interface for a container registry named 'pa427100003'. The page title is 'pa427100003 | Repositories'. Below the title, there are buttons for 'Refresh' and 'Manage Deleted Repositories'. A search bar is present with the placeholder text 'Search to filter repositories ...'. Below the search bar, there is a table with two columns: 'Repositories' and 'Cache Rule'. The table lists three repositories: 'func-app', 'report-job', and 'validate-api', each with a three-dot menu icon to its right.

Repositories	Cache Rule
func-app	...
report-job	...
validate-api	...

all 3 repos showing in the portal

TASK 3:

check function_app in repo

The screenshot shows the Azure Portal interface for a Function App. The top navigation bar includes 'Home', the function app name 'pa4-27100003-func', and various action buttons like 'Browse', 'Refresh', 'Stop', 'Restart', 'Swap', 'Get publish profile', 'Reset publish profile', 'Download app content', 'Delete', and 'Send us your feedback'. Below the navigation bar, the 'Essentials' tab is active, displaying key information about the function app. The 'Functions' tab is also visible, showing a list of functions.

Essentials	
Resource group (move)	: rg-sp26-27100003
Status	: Running
Location (move)	: UK West
Subscription (move)	: Microsoft Azure - Ali Khawaja
Subscription ID	: 67e93b84-fe08-452c-80ea-175d0a3eca56
Default domain	: pa4-27100003-func-htfjahqhef6gvee.ukwest-01.azurewe...
Health Check	: Not Configured
Operating System	: Linux
App Service Plan	: pa4-27100003 (B1: 1)
Runtime version	: 4.1048.200.3

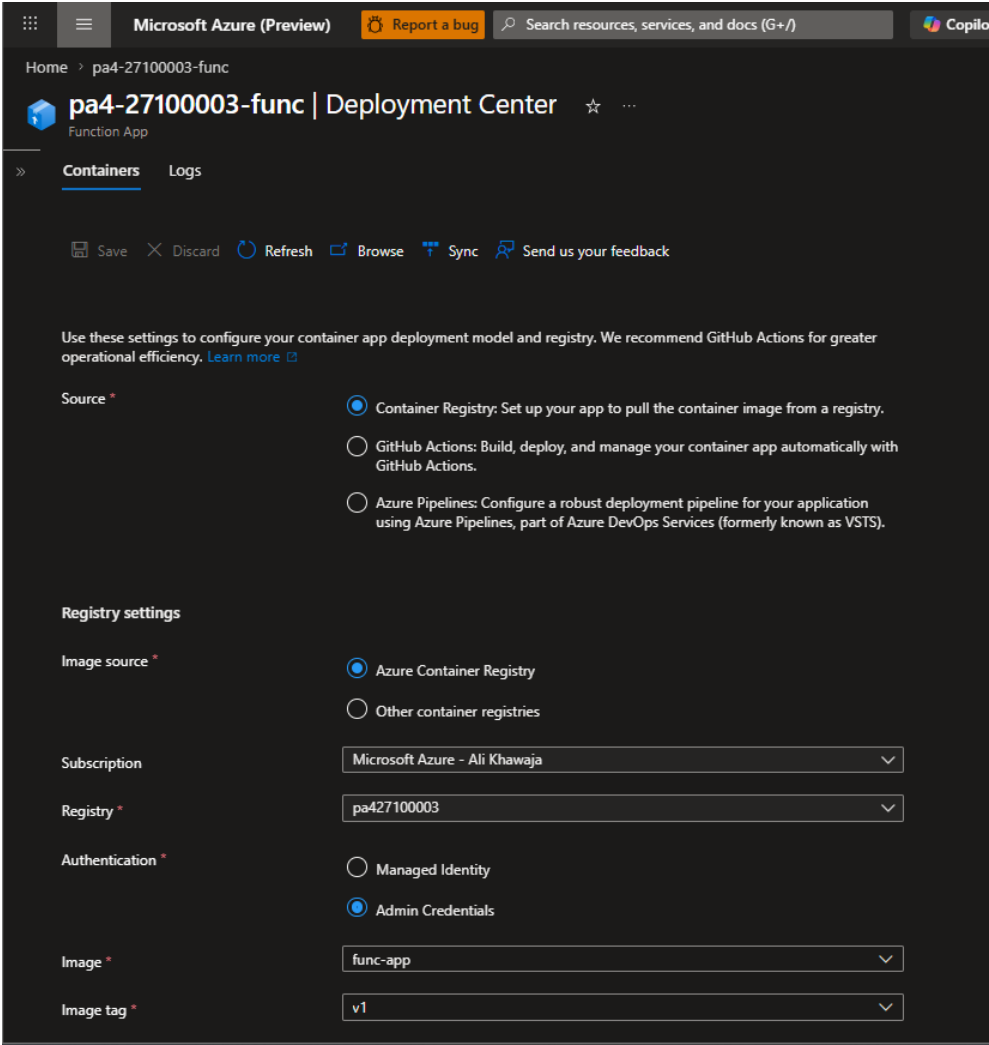
Functions			
Name	Trigger	Status	Monitor
http_starter	HTTP	Enabled	Invocations and more
my_orchestrator	Orchestration	Enabled	Orchestration information
report_activity	Activity	Enabled	Invocations and more
validate_activity	Activity	Enabled	Invocations and more

func start was not working at all; after a million attempts i just took a ss of function app lisitng the same 4 functions (same purpose)

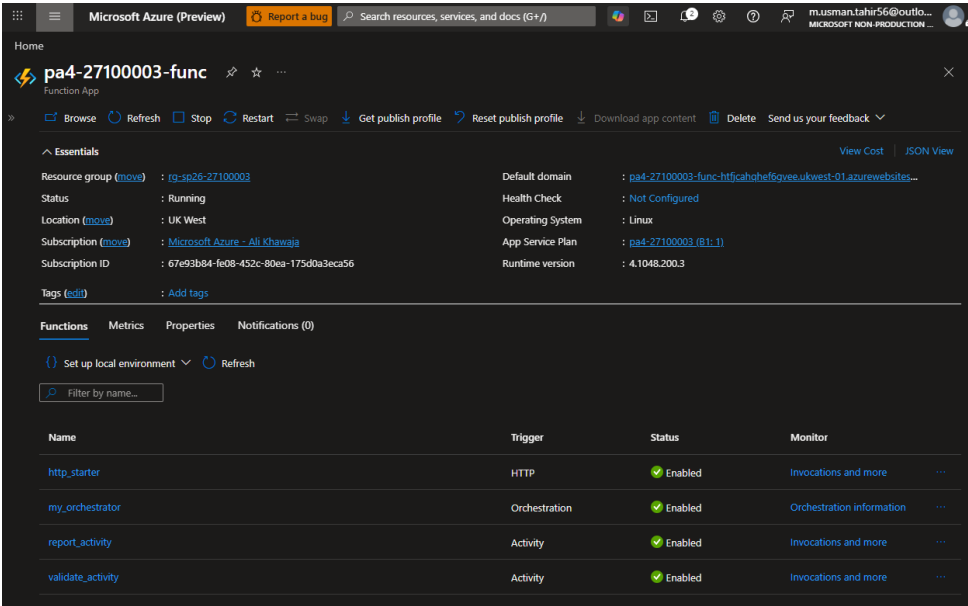
```
Hp@Usman MINGW64 /c/Spring 2026/Cloud Development/PA4-1/CS487-PA4 (main)
$ curl http://51.104.38.31:8080/validate \
  -X POST \
  -H "Content-Type: application/json" \
  -d '{"order_id":"TEST-001","items":[{"sku":"ABC","qty":2}]}'\
  {"valid":true,"reason":"ok","order_id":"TEST-001"}
```

Local smoke test of validate_activity against AKS

TASK 4:



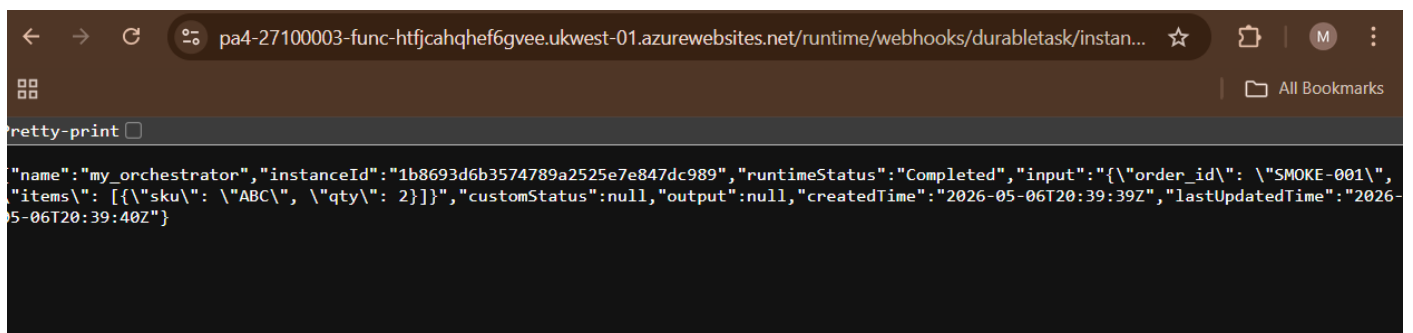
Function App Overview showing container image configuration



all 4 functions showing in a list

```
>> curl.exe -X POST "https://pa4-27100003-func-htfjcahqhef6gvee.ukwest-01.azurewebsites.net/api/orchestrators/my_orchestra
tor?code=OfrbjoES72ZpgeGkPn_rwRiR1p606wCT6wpV_d9ELAsAzFuvq1eJw==" -H "Content-Type: application/json" -d $body
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> $body = '{"order_id": "SMOKE-001", "items": [{"sku": "ABC", "qty": 2}
]}'
>>
>> $response = Invoke-WebRequest -Uri "https://pa4-27100003-func-htfjcahqhef6gvee.ukwest-01.azurewebsites.net/api/orchestr
ators/my_orchestrator?code=OfrbjoES72ZpgeGkPn_rwRiR1p606wCT6wpV_d9ELAsAzFuvq1eJw==" -Method POST -ContentType "applicatio
n/json" -Body $body -UseBasicParsing
>>
>> $response.Content
{"id": "1b8693d6b3574789a2525e7e847dc989", "statusQueryGetUri": "https://pa4-27100003-func-htfjcahqhef6gvee.ukwest-01.azur
ebsites.net/runtime/webhooks/durabletask/instances/1b8693d6b3574789a2525e7e847dc989?taskHub=pa427100003func&connection=S
torage&code=XcwVljOvaSRgkvW7tWTz63VLOrqlOku5fporvTA6DumAzFuusCkGQ==", "sendEventPostUri": "https://pa4-27100003-func-htfj
cahqhef6gvee.ukwest-01.azurewebsites.net/runtime/webhooks/durabletask/instances/1b8693d6b3574789a2525e7e847dc989/raiseEven
t/{eventName}?taskHub=pa427100003func&connection=Storage&code=XcwVljOvaSRgkvW7tWTz63VLOrqlOku5fporvTA6DumAzFuusCkGQ==", "
terminatePostUri": "https://pa4-27100003-func-htfjcahqhef6gvee.ukwest-01.azurewebsites.net/runtime/webhooks/durabletask/in
stances/1b8693d6b3574789a2525e7e847dc989/terminate?reason={text}&taskHub=pa427100003func&connection=Storage&code=XcwVljOva
SRgkvW7tWTz63VLOrqlOku5fporvTA6DumAzFuusCkGQ==", "rewindPostUri": "https://pa4-27100003-func-htfjcahqhef6gvee.ukwest-01.a
zurewebsites.net/runtime/webhooks/durabletask/instances/1b8693d6b3574789a2525e7e847dc989/rewind?reason={text}&taskHub=pa42
7100003func&connection=Storage&code=XcwVljOvaSRgkvW7tWTz63VLOrqlOku5fporvTA6DumAzFuusCkGQ==", "purgeHistoryDeleteUri": "h
ttps://pa4-27100003-func-htfjcahqhef6gvee.ukwest-01.azurewebsites.net/runtime/webhooks/durabletask/instances/1b8693d6b3574
789a2525e7e847dc989?taskHub=pa427100003func&connection=Storage&code=XcwVljOvaSRgkvW7tWTz63VLOrqlOku5fporvTA6DumAzFuusCkGQ
==", "restartPostUri": "https://pa4-27100003-func-htfjcahqhef6gvee.ukwest-01.azurewebsites.net/runtime/webhooks/durabletas
k/instances/1b8693d6b3574789a2525e7e847dc989/restart?taskHub=pa427100003func&connection=Storage&code=XcwVljOvaSRgkvW7tWTz6
3VLOrqlOku5fporvTA6DumAzFuusCkGQ==", "suspendPostUri": "https://pa4-27100003-func-htfjcahqhef6gvee.ukwest-01.azurewebsite
s.net/runtime/webhooks/durabletask/instances/1b8693d6b3574789a2525e7e847dc989/suspend?reason={text}&taskHub=pa427100003fun
c&connection=Storage&code=XcwVljOvaSRgkvW7tWTz63VLOrqlOku5fporvTA6DumAzFuusCkGQ==", "resumePostUri": "https://pa4-2710000
3-func-htfjcahqhef6gvee.ukwest-01.azurewebsites.net/runtime/webhooks/durabletask/instances/1b8693d6b3574789a2525e7e847dc98
9/resume?reason={text}&taskHub=pa427100003func&connection=Storage&code=XcwVljOvaSRgkvW7tWTz63VLOrqlOku5fporvTA6DumAzFuusC
kGQ=="}
```

curl/JSON output showing orchestration started (instance id + statusQueryGetUri)



Screenshot of the status query URL JSON showing Completed not failed because i did task 5 and 6 first

TASK 5:

```
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> az aks get-credentials --resource-group rg
-sp26-27100003 --name pa4-27100003
Merged "pa4-27100003" as current context in C:\Users\Hp\.kube\config
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
aks-nodepool1-28032298-vmss000000	Ready	<none>	4m32s	v1.34.6

node is ready

```
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
validate-deployment-7dfd7995f8-8h7d7	1/1	Running	0	18m

ss of get pods running successfully

```
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> kubectl get service
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.0.0.1	<none>	443/TCP	35m
validate-service	LoadBalancer	10.0.62.206	51.104.38.31	8080:32517/TCP	27m

assigned external ip

```
PS C:\Spring 2026\Cloud Development\PA4-1\CS487-PA4> curl http://51.104.38.31:8080/health
Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): y

StatusCode      : 200
StatusDescription : OK
Content         : {"status":"ok"}
RawContent      : HTTP/1.1 200 OK
                  Content-Length: 15
                  Content-Type: application/json
                  Date: Wed, 06 May 2026 18:37:09 GMT
                  Server: uvicorn

                  {"status":"ok"}
Forms           : {}
Headers         : {[Content-Length, 15], [Content-Type, application/json], [Date, Wed, 06
                  May 2026 18:37:09 GMT], [Server, uvicorn]}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
```

curl health check

The screenshot shows the Azure portal interface for a Function App named 'pa4-27100003-func'. The 'Environment variables' tab is selected, displaying a table of variables. The 'VALIDATE_URL' variable is highlighted, showing its value as 'http://51.104.38.31:8080/validate'.

Name	Value	Deployment slot setting	Source
APPLICATIONINSIGHTS_CON...	Show value		App Service
AzureWebJobsStorage	Show value		App Service
DOCKER_REGISTRY_SERVER_P...	Show value		App Service
DOCKER_REGISTRY_SERVER_U...	Show value		App Service
DOCKER_REGISTRY_SERVER_U...	Show value		App Service
FUNCTIONS_EXTENSION_VER...	Show value		App Service
VALIDATE_URL	http://51.104.38.31:8080/validate		App Service
WEBSITES_ENABLE_APP_SERVI...	Show value		App Service

showing value of VALIDATE_URL

TASK 6:

```

Hp@Usman MINGW64 /c/Spring 2026/Cloud Development/PA4-1/CS487-PA4 (main)
$ az container create \
  --resource-group rg-sp26-27100003 \
  --name ci-report-test \
  --image pa427100003.azurecr.io/report-job:v1 \ ...
  "clientId": "9007f99d-03c8-425b-b991-4ba68769fe6f",
  "principalId": "04ef8e41-8654-4387-88b0-7c2d2e726eb2"
}
},
"imageRegistryCredentials": [
  {
    "identity": null,
    "identityUrl": null,
    "isDelegatedIdentity": false,
    "password": null,
    "server": "pa427100003.azurecr.io",
    "username": "pa427100003"
  }
],
"initContainers": [],
"instanceView": {
  "events": [],
  "state": "Succeeded"
},
"ipAddress": null,
"isCreatedFromStandbyPool": false,
"location": "ukwest",
"name": "ci-report-test",
"osType": "Linux",
"priority": null,
"provisioningState": "Succeeded",
"resourceGroup": "rg-sp26-27100003",
"restartPolicy": "Never",
"sku": "Standard",
"standbyPoolProfile": null,
"subnetIds": null,
"tags": {},
"type": "Microsoft.ContainerInstance/containerGroups",
"volumes": null,
"zones": null

```

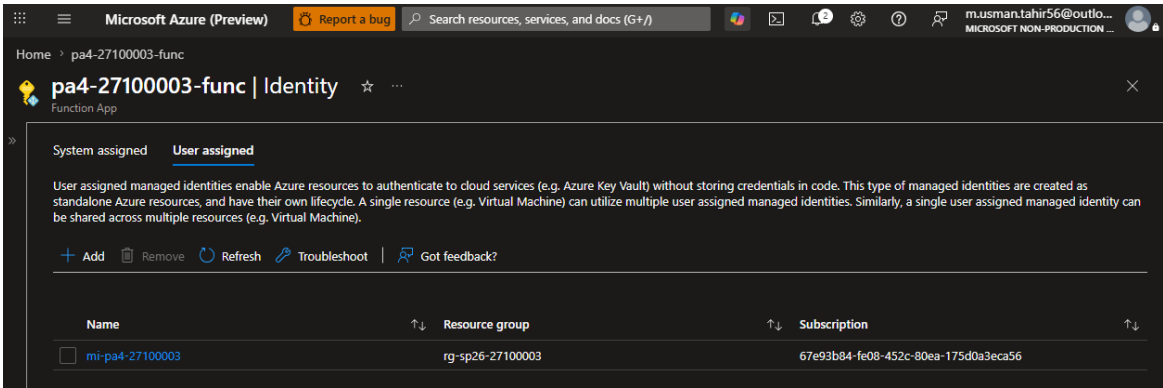
state: Succeeded

```
Hp@Usman MINGW64 /c/Spring 2026/Cloud Development/PA4-1/CS487-PA4 (main)
$ az container logs --resource-group rg-sp26-27100003 --name ci-report-test
Uploaded TEST-001.pdf to reports container
```

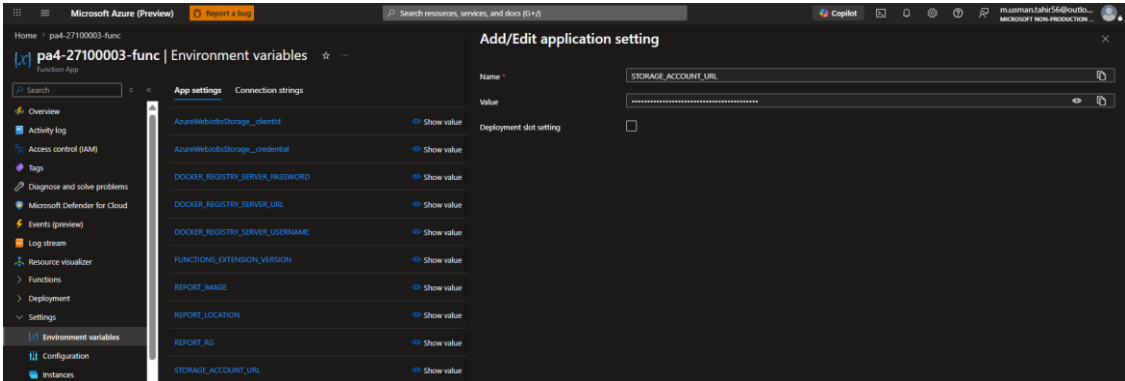
uploaded pdf successfully

```
Hp@Usman MINGW64 /c/Spring 2026/Cloud Development/PA4-1/CS487-PA4 (main)
$ az storage blob list --account-name pa427100003 --container-name reports --auth-mode login --output table
Name                               Blob Type  Blob Tier  Length  Content Type  Last Modified  Snapshot
-----
TEST-001.pdf                       BlockBlob  Hot        1462    application/octet-stream  2026-05-06T20:54:02+00:00
```

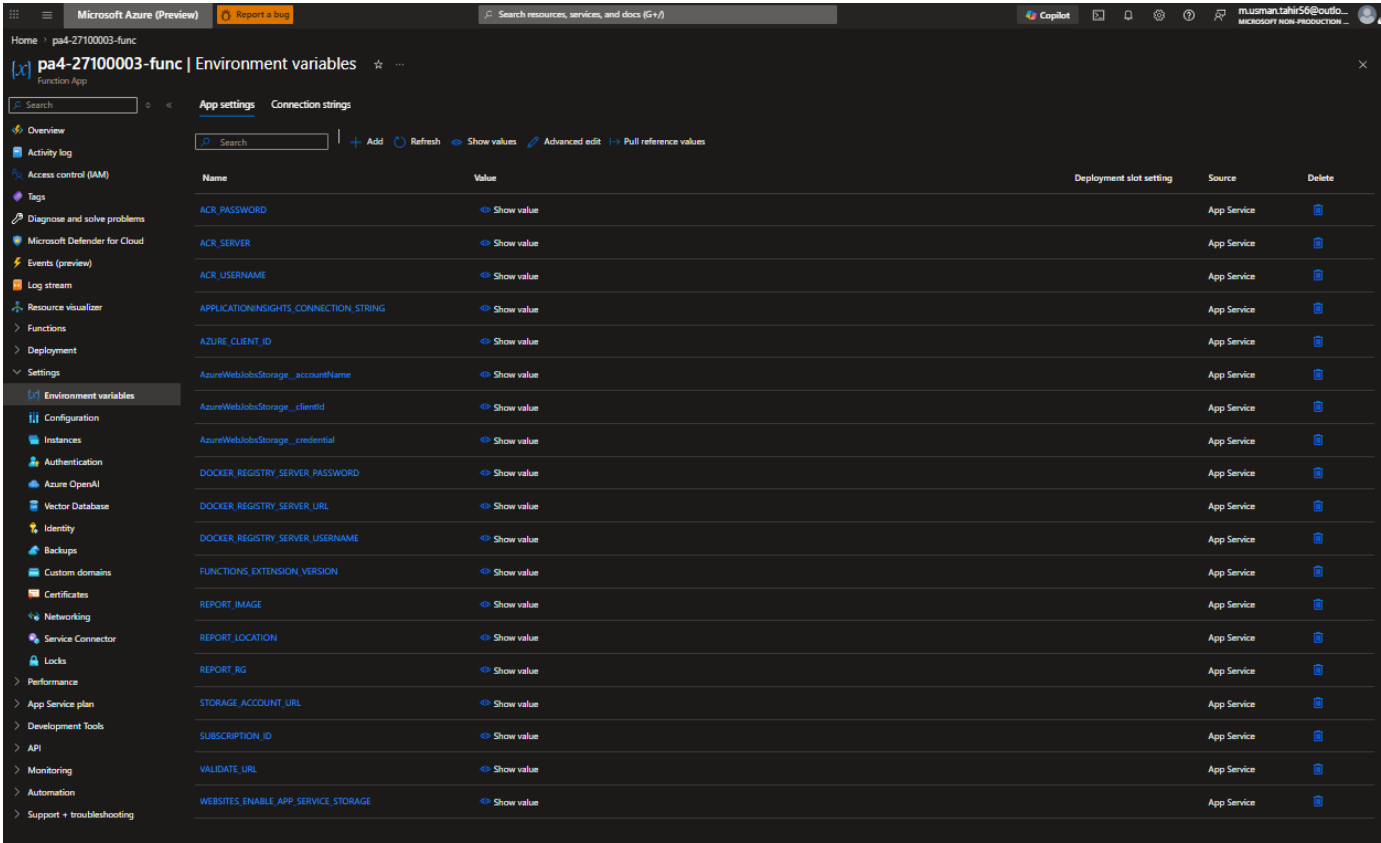
verified: pdf is in blob storage



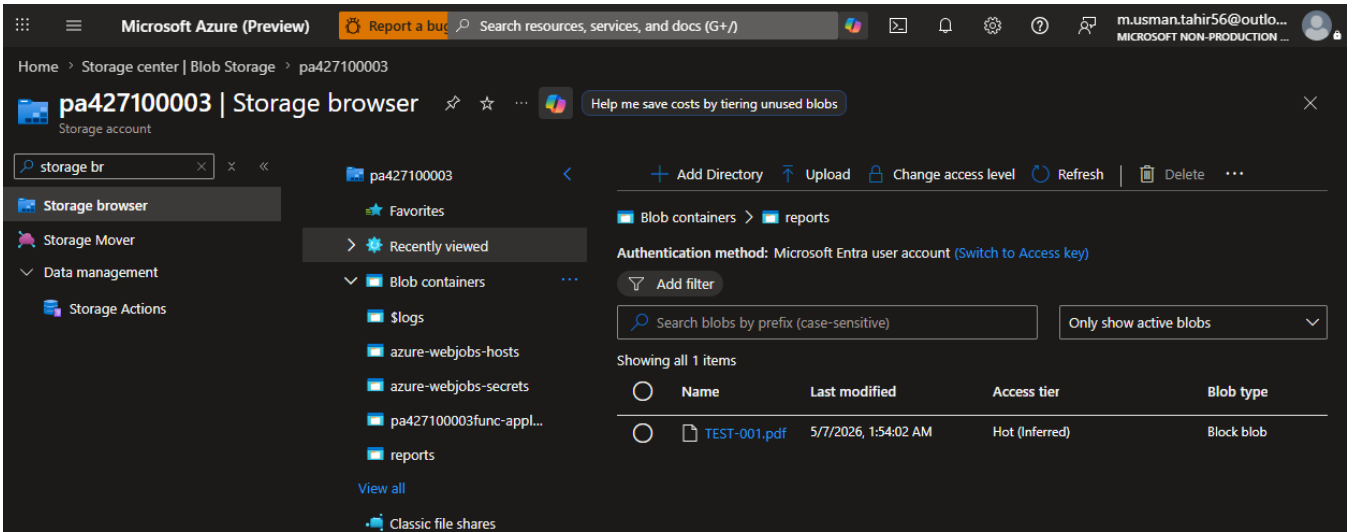
User assigned showing mi-pa4-27100003 attached



STORAGE_ACCOUNT_URL



Environment variables showing REPORT_*, ACR_*, STORAGE_ACCOUNT_URL values



pdf listed

TASK 7:

The screenshot shows a web browser window with the address bar displaying "pa4-27100003.azurewebsites.net". The page features a dark blue background with a light blue cloud pattern. In the center, there is a dark blue rounded rectangle containing the "TaskFlow" title. Below the title, there are three input fields: "Order ID" with the value "ORD-001", "SKU" with the value "WIDGET-X", and "Quantity (Max 100)" with the value "2". A purple "Submit Order" button is positioned below these fields. At the bottom of the form, a green "Completed" status message is displayed, followed by the text "No report URL was returned."

TaskFlow

Order ID

ORD-001

SKU

WIDGET-X

Quantity (Max 100)

2

Submit Order

Completed

No report URL was returned.

qty =2

The screenshot shows the same web browser window as before, but with the "Quantity" input field set to "0". The "Submit Order" button is still present. Below the button, a red "Invalid Input" error message is displayed, stating "Enter an order ID, SKU, and quantity of at least 1."

TaskFlow

Order ID

ORD-001

SKU

WIDGET-X

Quantity (Max 100)

0

Submit Order

Invalid Input Enter an order ID, SKU, and quantity of at least 1.

invalid input

TASK 8:

CHECK SUBMISSION.MD